

Optimizing quasisymmetry with meow

From boundary coefficients
to a quasisymmetric equilibrium.

C++20 · TRF

cuMES tangents

Landreman QA / QH

DENSE OPTIMIZATION PROBLEM

120

boundary variables at QH mode 5

MEASURED FULL CONSTRUCTION SPEED

$2.73 \times / 2.65 \times$

QA / QH · four cold FD workers + Broyden

Two questions for this presentation

01 · HOW THE OPTIMIZATION WORKS

Which work belongs to meow and which belongs to cuMES? How can an equilibrium tangent replace a nonlinear finite-difference solve?

Measure both execution time and the final objective.

02 · WHAT HAS BEEN REPRODUCED

Reconstruct QA and QH from analytic seeds. Check the published boundaries with an independent equilibrium solver.

Keep construction, final refinement, and high-resolution re-solves distinct.

meow closes the loop around cuMES

01

Boundary x

meow selects absolute Fourier coefficients and a fixed accepted-axis predictor.

02

Equilibrium $u(x)$

cuMES validates the fixed-boundary problem and solves its multigrid force balance on the GPU.

03

Residual $r(x)$

meow samples fields and profiles, then forms QS, aspect, and optional mean- ι residuals.

04

Jacobian J

Use cold differences, retained equilibrium tangents, or a safeguarded secant update.

Optimize the boundary, hold its scale

$$R = \sum_{m,n} R_{mn}^c \cos(m\theta - nn_{fp}\zeta), \quad Z = \sum_{m,n} Z_{mn}^s \sin(m\theta - nn_{fp}\zeta)$$

Maximum active mode k	1	2	3	4	5
Boundary variables: $4k(k+1)$	8	24	48	80	120

PARAMETERIZATION

Stellarator symmetry retains RBC and ZBS. At $m = 0$, keep only positive n ; for $m \geq 1$, keep $-k \leq n \leq k$.

SCALE AND RESOLUTION

RBC(0,0) stays fixed at 1. Optimizer cutoff k and the cuMES equilibrium basis are separate controls.

TRF turns residuals into boundary steps

$$\min_x \frac{1}{2} \|r(x)\|^2, \quad q(p) = \frac{1}{2} \|r + Jp\|^2, \quad \|D^{-1}p\| \leq \Delta$$

MODEL + PROPOSAL

Form the dense scaled Hessian from $J^T J$. Solve the trust-region quadratic; the general library also handles bounds with Coleman–Li scaling and reflected steps.

TRIAL + ACCEPTANCE

Compare actual and predicted reduction. Good trials update the boundary and axis predictor; poor trials shrink the radius. f_{tol} , x_{tol} , and g_{tol} stop the run.

A small step satisfies x_{tol} . It does not certify a good QS objective or an accurate Jacobian.

A Jacobian is the expensive part

BLACK-BOX FINITE DIFFERENCES

$$J_j \approx \frac{r(x + h_j e_j) - r(x)}{h_j}$$

Each column requires a perturbed, converged equilibrium and target evaluation. At mode 5: 120 perturbations per full refresh, plus the cached center.

EXTRA INFORMATION FROM CUMES

$$F_u u_j = -F_{x_j}, \quad J_j = T_{x_j} + T_u u_j$$

Retain the final-grid linearization. Solve a linear equilibrium response per boundary direction, then apply meow's analytic target chain rule.

The tangent crosses a clear API boundary

01

Retain

EquilibriumLinearization owns the converged force linearization and iterative solve session.

02

Differentiate

solve_boundary_tangent maps a BoundaryTangent to the spectral state response.

03

Materialize

materialize_tangent publishes field/profile derivatives; meow applies QS and size JVPs.

Current analytic selector	Policy at meow de1b324
Linear solve	relative tolerance 5×10^{-6} · absolute 10^{-11} · restart 300 · cap 1,000
Cold FD fallback	All $m = 0$ boundary directions; failed or nonfinite tangent columns
Status	Experimental hybrid Jacobian; cold finite differences remain the default

Tangents remove nonlinear work

Full QH construction · early experiment	Cold FD	Tangent
Accepted optimizer iterations	88	218
Equilibrium evaluations	4,303	253
Nonlinear equilibrium iterations	12,203,424	474,719
Tangent Jacobians	0	223
Tangent linear iterations	0	635,648

17.0× fewer equilibrium evaluations and 25.7× fewer nonlinear iterations. More optimizer steps and linear work remain.

Early speed comes with worse endpoints

Case	FD → tangent wall	Raw speed	FD objective	Tangent objective	Objective ratio
QA	1,537.26 → 473.29 s	3.248×	5.94684×10^{-7}	6.29151×10^{-4}	1,058.0×
QH	3,262.78 → 1,387.86 s	2.351×	5.36939×10^{-5}	1.28436×10^{-3}	23.9×

3.248× QA and 2.351× QH are raw wall-time ratios. Neither run qualifies as acceleration to the same optimization quality.

Check the whole Jacobian, then the endpoint

A DIRECTION CAN LOOK ACCURATE

After the cuMES correction, QH objective directional derivatives differ by at most 0.705% in the hybrid check.

But complete target-residual columns still differ by 2.95–11.68%. The 1% column gate is not met.

TRF USES MORE THAN $J^T r$

$$g = J^T r, \quad H \approx J^T J$$

A good directional objective derivative can hide a different curvature model. Tightening GMRES to 10^{-8} still leaves a 7.77% worst column difference.

Correct wiring changes the tangent result

CUMES DEFECT AND CORRECTION

The dual inverse transform wrote constraint fields into buffers that the constraint operator did not read. Commit **8c61395** connects the views.

Commit **f1a3ed7** initializes tangent de-aliasing scratch found by Compute Sanitizer.

MEASURED CORRECTED RESPONSE

Centered Solovev restart oracle	Difference
Aggregate spectral state	0.228%
Published equilibrium fields	0.514%

The harder QH columns still fail the agreed 1% target-column gate.

Corrected tangent: no qualified speedup yet

Case	Cold FD (s)	Corrected (s)	Speed / FD	Final objective	Objective / FD
QA	1,537.26	2,628.35	0.585×	1.34527×10^{-6}	2.26×
QH	3,262.78	3,516.16	0.928×	9.71168×10^{-5}	1.81×

Corrected endpoints improve greatly, but QA takes 1.71× the control time and QH takes 1.08×. Both objectives remain worse.

Reuse reliable derivatives between refreshes

$$s = x_{k+1} - x_k, \quad y = r_{k+1} - r_k, \quad J_{k+1} = J_k + \frac{(y - J_k s) s^T}{s^T s}$$

SAFEGUARDED BROYDEN UPDATE

An accepted step supplies a secant. Refresh from cold finite differences when the model becomes unreliable or the refresh interval is reached.

QUALIFIED AGGRESSIVE SETTINGS

Refresh interval	8 accepted steps
Minimum reduction ratio	0.05
Maximum relative secant error	0.5

Broyden uses already observed residual changes. It is separate from cuMES equilibrium-tangent information.

Schedule independent columns concurrently

Mode-1 Jacobian · 8 columns	Serial	2 workers	4 workers	4-worker matrix difference
QA	2.829 s	1.263 s	0.867 s	0 · bit-exact
QH	2.282 s	1.035 s	0.735 s	0 · bit-exact

INDEPENDENT CUMES INSTANCES

Each worker owns its solve and target evaluation. The center residual and accepted-axis predictor stay fixed across the batch.

MEASURED SATURATION

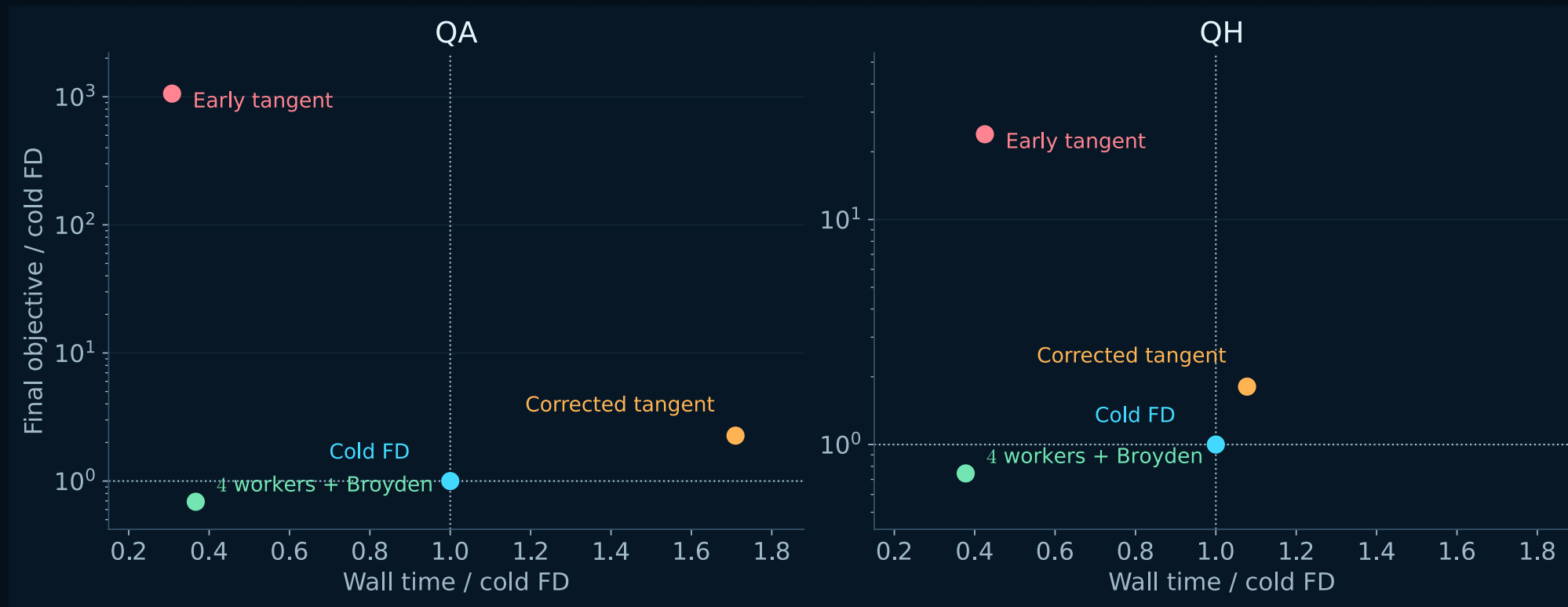
Eight workers: a QA perturbation lost convergence. QH stayed exact but slowed from 0.718 to 0.789 s in that separate diagnostic.

Qualified full constructions are faster

Jacobian strategy	QA wall (s)	QH wall (s)	End result
Historical cold FD	1,537.26	3,262.78	Control objectives
Serial aggressive Broyden	1,209.77	2,681.44	Same endpoints as both parallel Broyden runs
2 workers + Broyden	800.78	1,741.81	Same equilibrium / nonlinear work as 4 workers
4 workers + Broyden	563.79	1,232.17	QA 4.09895×10^{-7} · QH 3.99556×10^{-5}

Four workers give 1.420× QA / 1.414× QH over two workers, with matching trajectories. Versus historical cold FD: 2.727 × /2.648×, with improved objectives.

Time and quality must improve together



Lower-left is better. Point ratios come from the archived complete runs; no interpolation or timing-error bars are implied.

Reproducing Landreman QA and QH

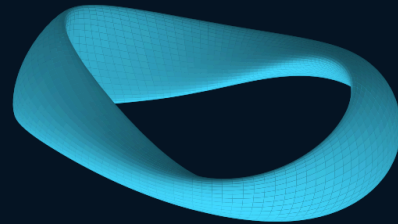
02

Use the original targets and seeds.

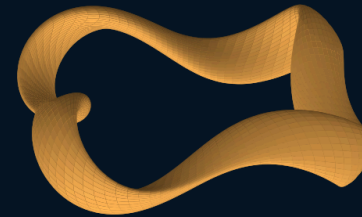
Inspect the shapes, convergence history, and independently solved fields.

The published geometries

QA · 2 field periods



QH · 4 field periods



Original published boundary coefficients · common geometric scale · shading represents surface geometry, not $|B|$.

QUASIAXISYMMETRY · QA

$n_{\text{fp}} = 2$ · target aspect ratio 6

Symmetric Boozer harmonics have $n = 0$.

QUASIHEDICAL SYMMETRY · QH

ZENODO V2 · CONFIGURATIONS/NEW_QA AND CONFIGURATIONS/NEW_QH · GENERATED FOURIER SURFACES

MEOW × CUMES

$n_{\text{fp}} = 4$ · target aspect ratio 8

Three distinct reproduction comparisons

Comparison	Starting data	What it establishes
Analytic-seed construction	Runs 021 (QA) / 039 (QH)	meow can rediscover low-QS shapes through mode continuation
Final-refinement target oracle	Terminal optimization wouts $\cdot n_s = 75$	The independent target evaluator matches saved SIMSOPT objectives
Published-boundary re-solve	configurations/new_QA / new_QH $\cdot n_s = 201$	cuMES solves the published geometry; compare its fields and QS profile

qa_analytic.json → construction. qa_start.json → final refinement. qa.json → published-boundary evaluation. QH uses the corresponding three inputs.

Match the physical QS functional

$$Q = \frac{(N - \iota M)(\mathbf{B} \times \nabla \psi) \cdot \nabla B + (MG + NI) \mathbf{B} \cdot \nabla B}{B^3}$$

$$r_{s,a} = \sqrt{w_s \frac{|\sqrt{g}_{s,a}|}{\sum_b |\sqrt{g}_{s,b}|}} Q_{s,a}, \quad f_{QS} = \sum_{s,a} r_{s,a}^2$$

FROM EQUILIBRIUM FIELDS

B is field strength. I and G are covariant flux functions; ψ uses the archived flux/sign convention. Normalize the surface quadrature once.

NO BOOZER SOLVE INSIDE OPTIMIZATION

Fourier-resample the cuMES fields and linearly interpolate half-grid quantities to the target surfaces. Boozer analysis comes afterward.

Keep targets and stages comparable

Setting	QA construction	QH construction	Final refinement
Aspect target	6	8	6/8
Mean- ι residual	$\bar{\iota} - 0.42$	Absent	Absent in both
Surface weights	All 1	All 1	QA: $1 \rightarrow 30$ · QH: $1 \rightarrow 2$
Target sampling	11 flux surfaces	$s = 0, 0.1, \dots, 1$	63×64 points / surface
FD relative / absolute step	$3.16228 \times 10^{-3} / 1 \times 10^{-7}$	$1 \times 10^{-3} / 1 \times 10^{-7}$	$1 \times 10^{-5} / 1 \times 10^{-9}$

44,352 QS residuals. Append aspect for both cases and mean ι only for QA construction. Objective = QS + scalar residual squares.

Unlock shape modes through continuation

Boundary cutoff k	Variables	cuMES $m_{\text{pol}}/n_{\text{tor}}$	QA	QH
1	8	3/3	Start	Start
2	24	5/5	Continue	Continue
3	48	6/6	Continue	Continue
4	80	6/6	Construction endpoint	Continue
5	120	6/6	Final refinement is separate	Construction endpoint

Use the imported $n_s = 12 \rightarrow 25 \rightarrow 50$ sequence for construction and a common 10^{-12} equilibrium gate. Carry the accepted axis when adding modes.

A faithful seed needs faithful derivative policy

QA STARTS ON A STATIONARY MANIFOLD

$$R = 1 + 0.2 \cos \theta, \quad Z = 0.2 \sin \theta$$

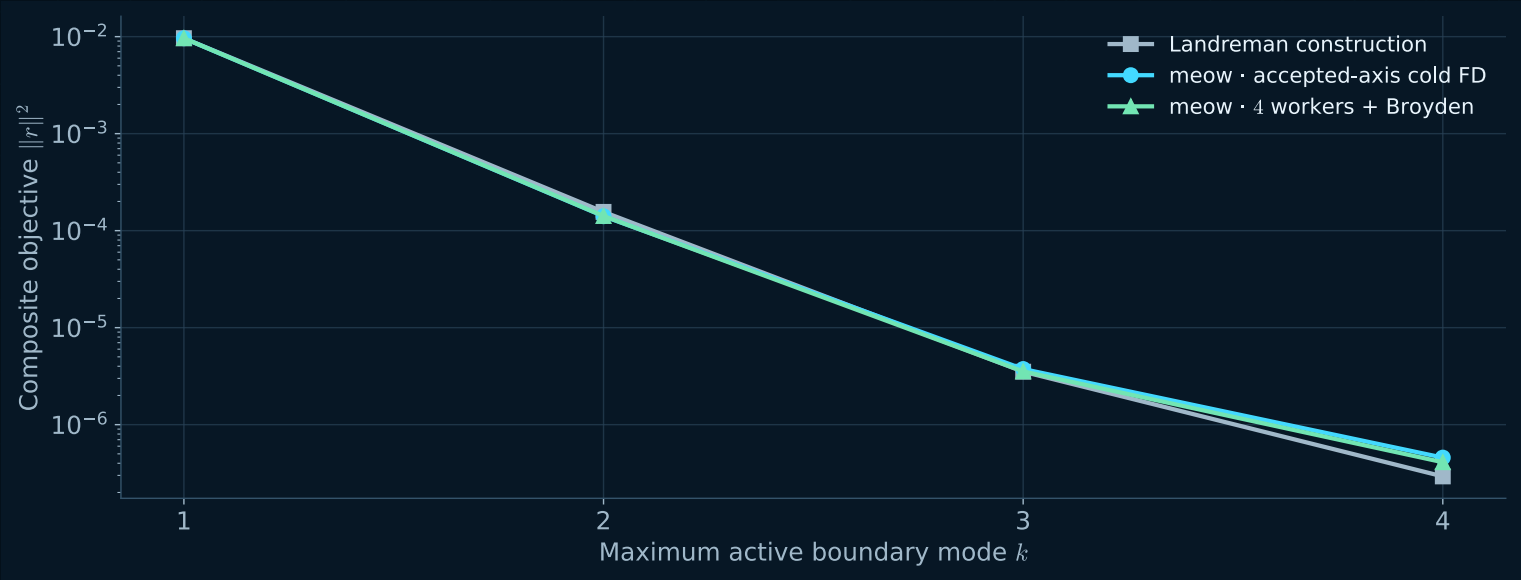
At exact axisymmetry the 3-D tangent cannot see first-order transform growth. The tangent path inserts a deterministic chiral seed of amplitude 10^{-4} .

COLD SOLVES NEED A CONSISTENT PREDICTOR

Reusing the refinement FD step left the QA objective at 0.0442 and mean ι near 0.284.

Construction steps plus accepted-axis tracking reach mode-1 objective 0.00962215; the archive gives ≈ 0.0096408 .

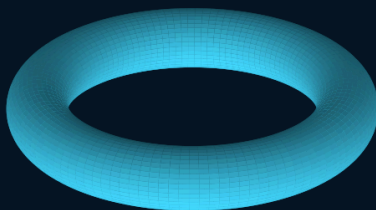
QA: construction reaches the reference scale



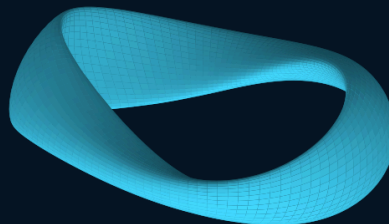
Final construction objective	Landreman	meow cold FD	meow 4-worker Broyden
$k = 4 \cdot$ same target definition	2.92794×10^{-7}	4.57544×10^{-7}	4.09895×10^{-7}
Ratio to Landreman	1.000	1.563	1.400

QA: the reconstructed boundary

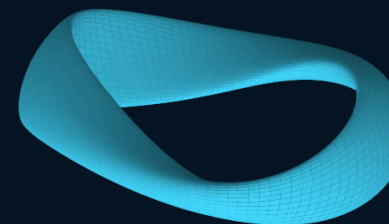
Analytic seed



meow · toroidal phase aligned



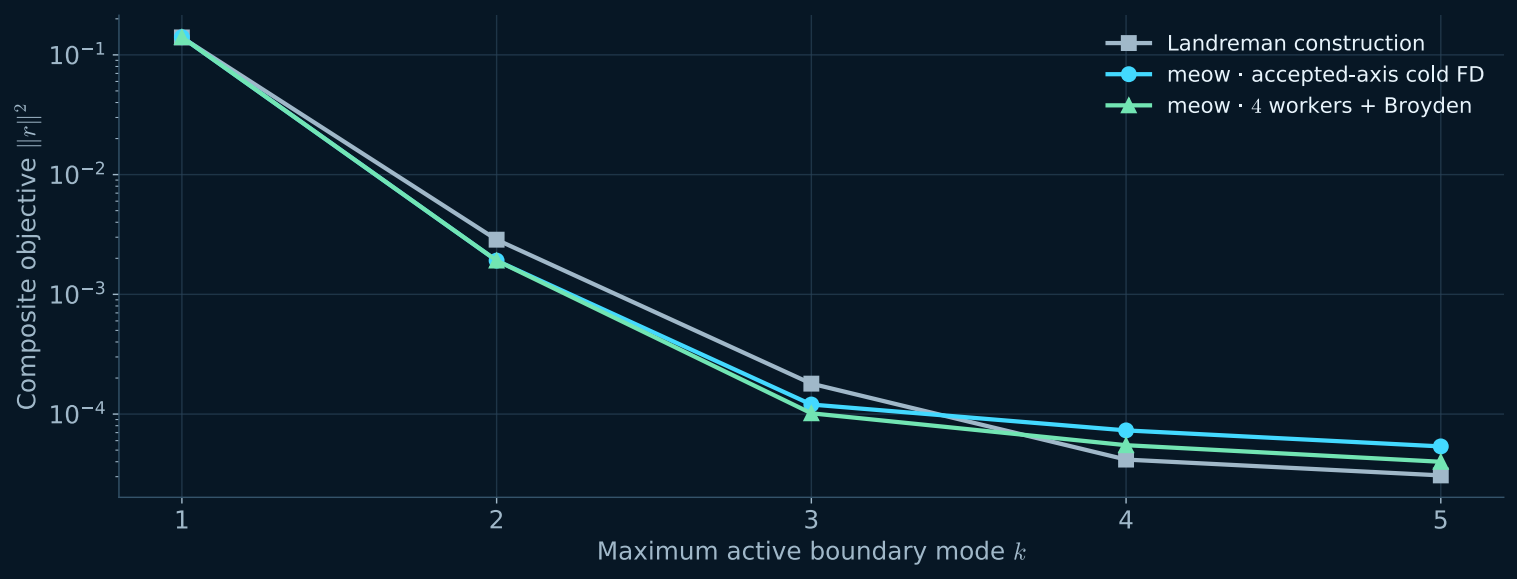
VMEC construction reference



Actual Fourier coefficients · common scale · QA rotated by π/n_{fp} to align toroidal phase · geometry shading only.

QA begins with an axisymmetric torus.

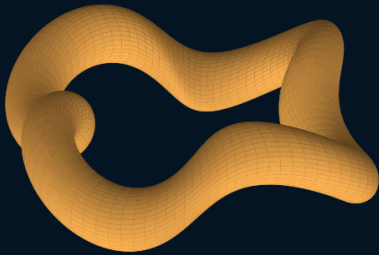
QH: construction reaches the reference scale



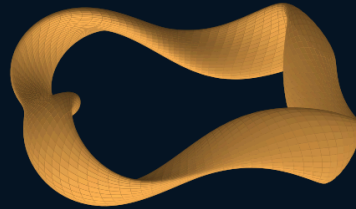
Final construction objective	Landreman	meow cold FD	meow 4-worker Broyden
$k = 5$ · same target definition	3.07486×10^{-5}	5.36939×10^{-5}	3.99556×10^{-5}
Ratio to Landreman	1.000	1.746	1.299

QH: the reconstructed boundary

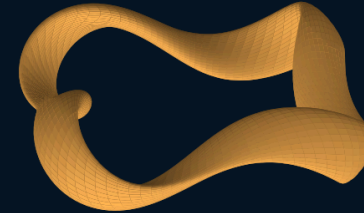
Analytic seed



meow construction



VMEC construction reference



Actual Fourier coefficients · common scale · QA rotated by π/n_{fp} to align toroidal phase · geometry shading only.

QH begins with $R = 1 + 0.2 \cos \theta + 0.2 \cos 4\zeta$ and $Z = 0.2 \sin \theta - 0.2 \sin 4\zeta$.

The target oracle reproduces the archived totals

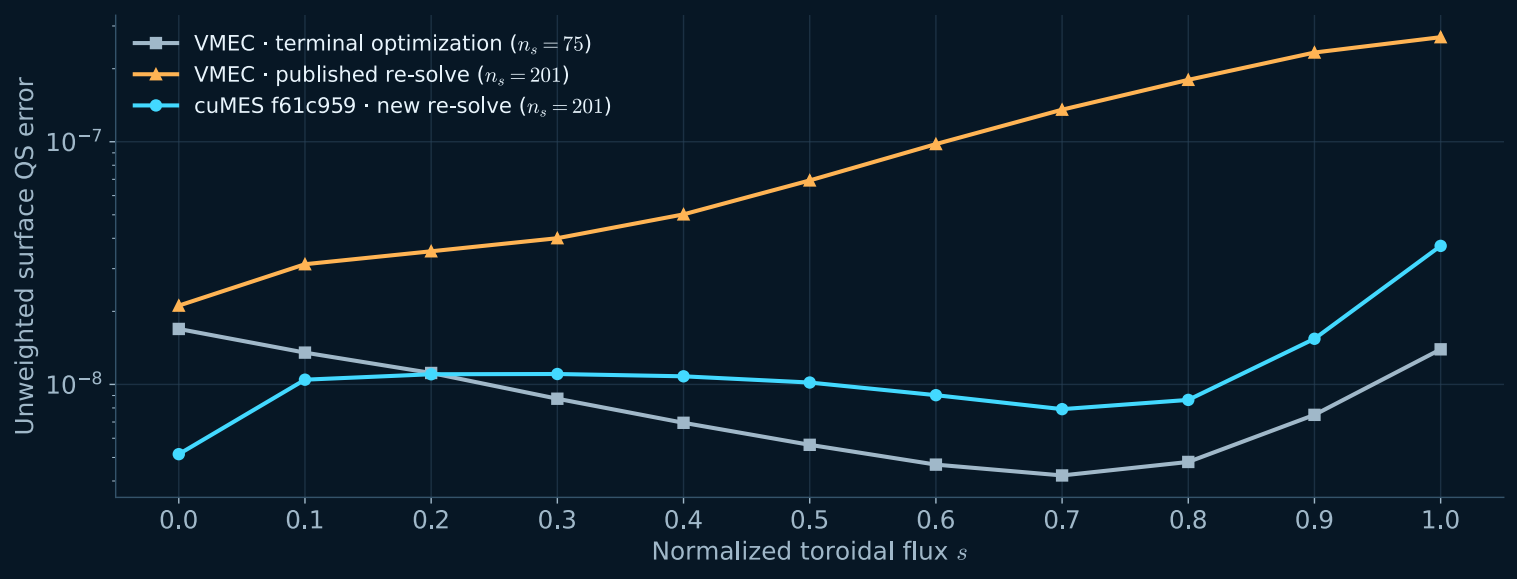
Terminal VMEC output · new CPU evaluation	QA	QH
Composite objective	$1.3183891708 \times 10^{-6}$	$3.4476418370 \times 10^{-5}$
Weighted QS	$1.3183284307 \times 10^{-6}$	$3.4476291444 \times 10^{-5}$
Unweighted QS	$9.8031606361 \times 10^{-8}$	$2.1107446594 \times 10^{-5}$
VMEC final grid	$n_s = 75 \cdot m_{\text{pol}} = 8 \cdot n_{\text{tor}} = 6$	$n_s = 75 \cdot m_{\text{pol}} = 8 \cdot n_{\text{tor}} = 6$

The independently evaluated terminal totals match the frozen SIMSOPT values to their printed precision. This verifies the target calculation on the same fields.

Published boundaries converge in cuMES

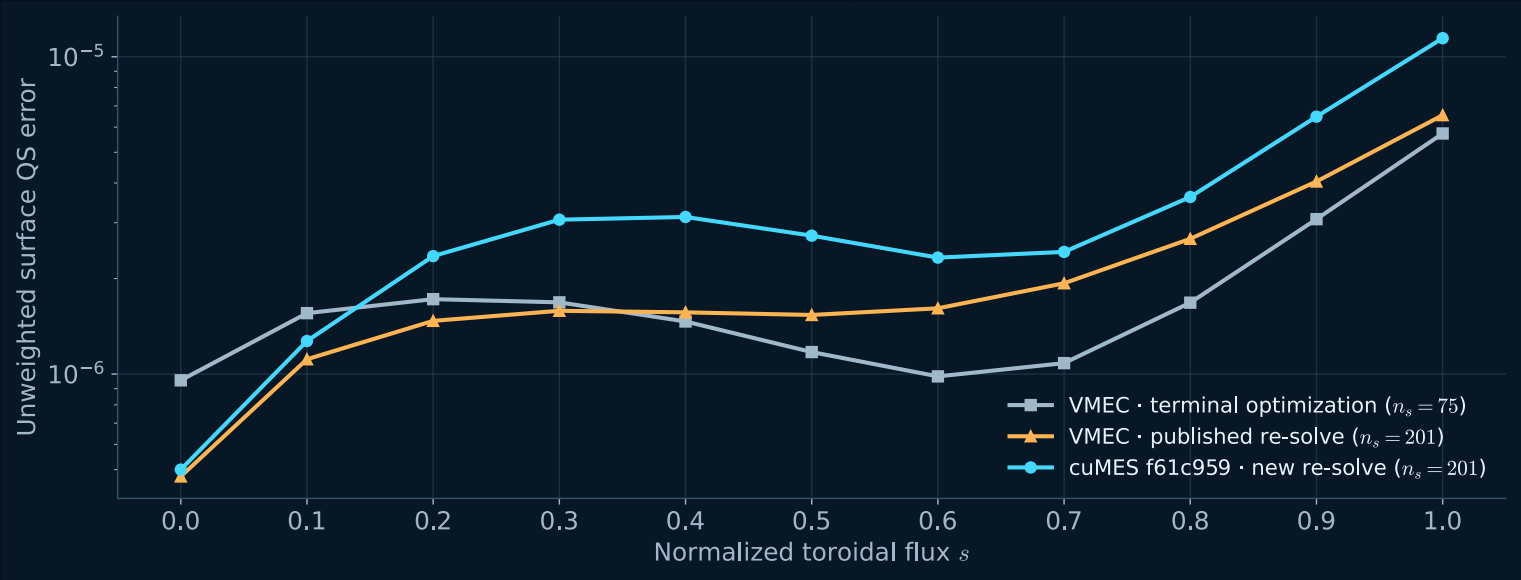
New GPU solve · 2026-09-08	QA	QH
Final resolution	$201 \times (m_{\text{pol}} = 16, n_{\text{tor}} = 12)$	$201 \times (m_{\text{pol}} = 8, n_{\text{tor}} = 8)$
Total / final-stage iterations	5,616/463	5,766/976
FSQR/FSQZ/FSQL	$9.99 \times 10^{-13} / 5.18 \times 10^{-13} / 9.36 \times 10^{-14}$	$9.97 \times 10^{-13} / 8.02 \times 10^{-13} / 1.54 \times 10^{-13}$
Aspect ratio	6.000007794	8.000011266
Mean ι	0.419325609	-1.243299682
Composite objective	2.60009×10^{-6}	6.67316×10^{-5}

QA: compare the full radial QS profile



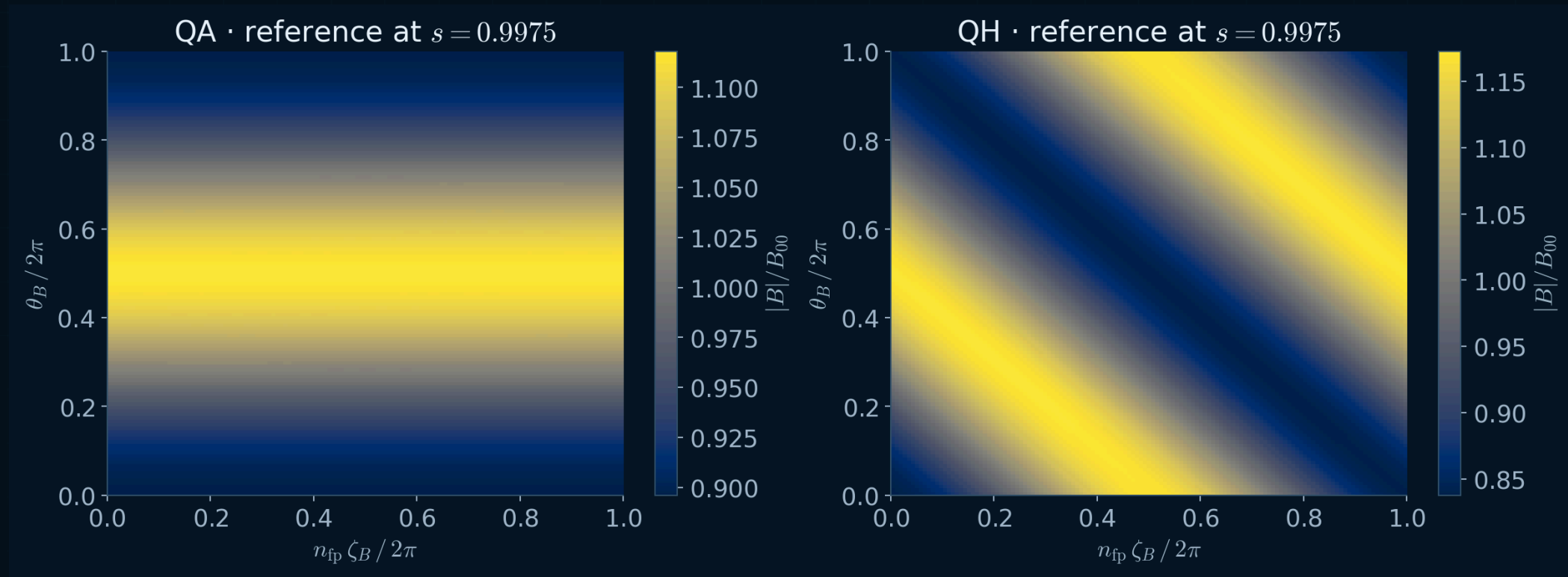
Composite objective	VMEC terminal · $n_s = 75$	VMEC published · $n_s = 201$	cuMES pinned · $n_s = 201$
QA	1.31839×10^{-6}	2.59293×10^{-5}	2.60009×10^{-6}

QH: compare the full radial QS profile



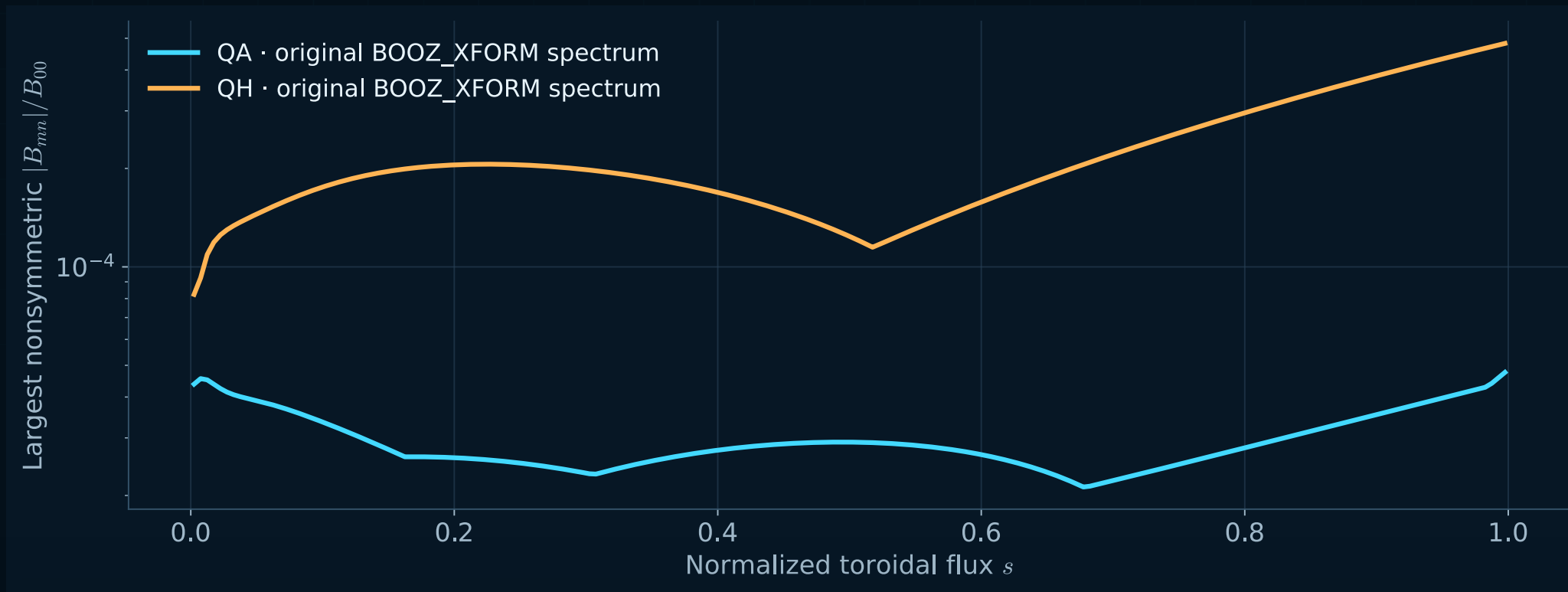
Composite objective	VMC terminal · $n_s = 75$	VMC published · $n_s = 201$	cuMES pinned · $n_s = 201$
QH	3.44764×10^{-5}	4.14606×10^{-5}	6.67316×10^{-5}

Boozer coordinates expose the reference symmetry



Original BOOZ_XFORM fields at $s = 0.9975$ · $|B|$ normalized by each surface's B_{00} · reference data, not new cuMES output.

A spectral diagnostic is a separate check



Reference maximum $|B_{mn}|/B_{00}$: QA 4.758×10^{-5} · QH 4.822×10^{-4} . These are spectral amplitudes, not the squared QS objective.

Run the current JSON workflow

PINNED INTEGRATION BUILD

```
cmake -S ../meow -B ../tmp/meow-slides-build \  
  -DMEOW_BUILD_CUMES_INTEGRATION=ON \  
  -DMEOW_FETCH_CUMES=ON \  
  -DCMAKE_BUILD_TYPE=Release  
cmake --build ../tmp/meow-slides-build -j 6
```

INSPECT THE SUPPLIED FAST RUNDOWN

```
../tmp/meow-slides-build/cumes_landreman_optimize \  
  --dry-run \  
  slides/meow/data/inputs/qa-four-worker.rundown.json
```

Remove `--dry-run` to optimize; QH has its own rundown. The JSON explicitly sets `workers=4`, `refresh_interval=8`, and the `0.05/0.5` safeguards.

What the measurements establish

OPTIMIZATION MECHANICS AND SPEED

cuMES supplies a reusable equilibrium response; meow owns the target and Jacobian chain rule.

Tangents reduce nonlinear work, but corrected full runs have no qualified speedup. Four-worker Broyden delivers the measured fast constructions.

LANDREMAN REPRODUCTION

Analytic-seed constructions reach the archived objective scale and recover similar QA/QH shapes.

The target oracle matches the original totals. Published-boundary cuMES solves converge, with documented differences in their radial QS profiles.